

Floating Object Solver v2 $\rightarrow$ v3

Self-consistent body dynamics via a monolithic constraint solve
(slides updated for v3 corrections per Galeano-Rios's April 2026 feedback)

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LNS Floating Project

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Outline

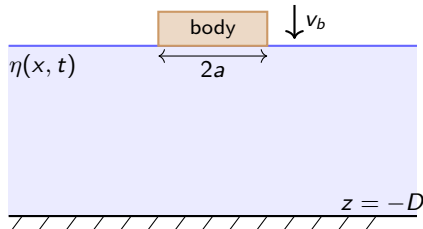
1. Motivation: from prescribed motion to self-consistent dynamics
2. Physical model: Helmholtz decomposition + body coupling
3. Monolithic numerical formulation
4. Python implementation
5. Concrete examples and validation
6. Limitations and future work

Goal: every equation traceable to `floating_object_v2.py`.

Problem setup

- ▶ 2D domain, periodic in x
- ▶ Viscous fluid below, free surface above
- ▶ **Flat-bottomed rigid body** sits on surface
- ▶ Body has half-width a , mass m (per unit length)
- ▶ Non-slip at bottom, $z = -D$
- ▶ At $t = 0$: body in contact with surface

Output we want: body trajectory $z_b(t)$, body velocity $v_b(t)$, surface elevation $\eta(x, t)$, surface pressure $p_s(x, t)$ — all computed self-consistently.



v1 vs v2 at a glance

	v1 (floating_object.py)	v2 (floating_object_v2.py)
Body motion	Prescribed: $z_b(t) = -\text{draft} + A \sin(\omega t)$	Computed: $M\ddot{z}_b = -M/\text{Fr} + \int p_s \, dx$
Role of body	Wavemaker (forced oscillator)	Rigid body with real dynamics
Surface pressure p_s	Not computed (ad-hoc)	Lagrange multiplier enforcing no-penetration
Free-surface BC under body	Overwrite $\eta \leftarrow z_b(t)$	Geometric constraint + kinematic match + horizontal no-slip
Unknowns per step	η, ϕ, w^3, w^1	$\eta, \phi, w^3, w^1, p_s, z_b, v_b$
Instability risk	Stable (forced)	Fully implicit $\Rightarrow$ no added-mass issues

Key insight

All new equations are *linear with constant coefficients* $\Rightarrow$ the monolithic prefactorisation trick carries over.

Helmholtz decomposition of the velocity field

Idea (Galeano-Rios, Milewski, Vanden-Broeck, JFM 2017)

Split the velocity field into irrotational + vortical parts:

$$\mathbf{u}(x, z, t) = \nabla\phi(x, z, t) + \mathbf{w}(x, z, t), \quad \nabla \cdot \mathbf{w} = 0.$$

- ▶ ϕ : velocity potential. Potential flow $\Rightarrow \Delta\phi = 0$ in the bulk.
- ▶ $\mathbf{w} = (w^1, w^3)$ in 2D; $w^2 = 0$ by symmetry.
- ▶ Each component of $\mathbf{w}$ diffuses: $\partial_t w^i = \nu \Delta w^i$.
- ▶ Coupling between the two fields happens at boundaries (surface, bottom).

Why: bulk dynamics dominated by potential flow; viscous effects concentrated near boundaries $\Rightarrow$ efficient linear PDE system instead of full nonlinear Navier–Stokes.

Bulk equations (dimensionless)

In the fluid domain $-D \leq z \leq 0$:

Potential

$$\Delta\phi = 0 \quad (\text{Laplace's equation — eq. 2.15a})$$

Vortical components

$$\partial_t w^i = \frac{1}{\text{Re}} \Delta w^i, \quad i = 1, 3 \quad (\text{heat equations — eq. 2.15b})$$

- ▶ $\Delta = \text{full 2D Laplacian } \partial_{xx} + \partial_{zz}$
- ▶ $\Delta_H = \partial_{xx} = \text{horizontal Laplacian (surface operator)}$
- ▶ Reynolds number $\text{Re} = UL_c/\nu$ controls viscous diffusion

Free-surface boundary conditions (outside body)

At $z = 0$, for $x \notin \text{body}$:

$$\partial_t \eta = \partial_z \phi + w^3 \quad (2.15c: \text{kinematic}) \quad (1)$$

$$\partial_t \phi_s = -\frac{\eta}{\text{Fr}} + \frac{1}{\text{We}} \kappa[\eta] + \frac{2}{\text{Re}} \Delta_H \phi_s - \frac{2}{\text{Re}} \partial_z w^3 \quad (2.15d: \text{dynamic, } p_s = 0) \quad (2)$$

$$\partial_t w^3 = \frac{2}{\text{Re}} \Delta_H (\partial_z \phi + w^3) \quad (2.15e: \text{vorticity}) \quad (3)$$

$$\partial_z w^1 = -(\partial_{xz} \phi + \partial_x w^3) \quad (2.11a: \text{stress-free}) \quad (4)$$

$\phi_s(x, t) \equiv \phi(x, 0, t)$ is the surface value of the potential; $\kappa[\eta] \approx \partial_{xx} \eta$ is the linearised curvature.

Remark

In this v2 implementation, **equation (2.15e) is solved directly** — we do *not* use the simplified diagnostic $w^3 = \frac{2}{\text{Re}} \Delta_H \eta$.

Under the body: three new constraints

Let $\mathcal{B}(t) = \{x : |x - x_c| < a\} = \text{wetted region}$.

(1) Geometric constraint — replaces free-surface kinematic

$$\eta(x, t) = z_b(t) \quad \forall x \in \mathcal{B}$$

the fluid surface under the body follows the body position.

(2) Kinematic match — no-penetration at body bottom

$$\partial_z \phi + w^3 = v_b(t) \quad \forall x \in \mathcal{B}, z = 0$$

vertical fluid velocity = body velocity (consistent with $\dot{\eta} = v_b$).

(3) Horizontal no-slip — replaces stress-free (2.11a)

$$\partial_x \phi + w^1 = 0 \quad \forall x \in \mathcal{B}, z = 0$$

horizontal fluid velocity vanishes at the rigid body bottom.

The dynamic BC (2.15d) **still holds everywhere**, but p_s is unknown under $\mathcal{B}$.

Surface pressure p_s : the Lagrange multiplier

Outside body: $p_s = 0$ (prescribed — free surface).

Under body: p_s becomes an **unknown**, determined by enforcing the no-penetration constraint $\partial_z \phi + w^3 = v_b$.

Quantity	Free surface ($x \notin \mathcal{B}$)	Wetted body ($x \in \mathcal{B}$)
η evolution	$\partial_t \eta = \partial_z \phi + w^3$	$\eta = z_b$
Kinematic	(evolves η)	$\partial_z \phi + w^3 = v_b$
Horizontal w^1	$2\partial_{xz} \phi + \partial_x w^3 + \partial_z w^1 = 0$ [v3 fix]	$\partial_x \phi + w^1 = 0$
Surface pressure	$p_s = 0$	$p_s = \text{unknown}$

Interpretation: p_s on $\mathcal{B}$ acts like a Lagrange multiplier — whatever pressure is required to make the fluid follow the body.

Two equivalent formulations of the body coupling

The coupled fluid–body problem can be written in either of two equivalent ways:

Formulation A — Lagrange multiplier (used in v3 code)

- ▶ State vector includes p_s as an explicit unknown: $\mathbf{x} = [w^3, w^1, \phi, \eta, p_s, z_b, v_b]$.
- ▶ p_s -rows of the matrix enforce $p_s = 0$ *outside body* and $\phi_z + w^3 = v_b$ *under body*.
- ▶ Newton's row: $M\dot{v}_b = -M/\text{Fr} + \int_B p_s \, dx$.
- ▶ Pro: p_s is a direct diagnostic output. Con: state vector has n_x extra rows.

Formulation B — Explicit traction (prof's main.tex)

- ▶ No p_s variable.
- ▶ Constraint $\phi_z + w^3 = v_b$ enforced as a direct row replacing the η -row under body.
- ▶ Newton's row uses the closed-form traction

$$\tau_z = -\partial_t \phi - \frac{\eta}{\text{Fr}} - \frac{2}{\text{Re}} \partial_z (\phi_z + w^3)$$

substituted explicitly into the integral.

- ▶ Pro: smaller system. Con: pressure must be reconstructed post-hoc.

Body dynamics: Newton's 2nd law

The body has scalar state (z_b, v_b) . Dimensionless evolution:

Kinematic

$$\frac{dz_b}{dt} = v_b$$

Newton's second law

$$M \frac{dv_b}{dt} = -\frac{M}{Fr} + \underbrace{\int_{\mathcal{B}} p_s(x, t) dx}_{\text{integrated fluid force}}$$

- ▶ $M = m/(\rho L_c^2)$: dimensionless mass (2D, per unit out-of-plane length)
- ▶ $-M/Fr$: gravity (weight)
- ▶ $+\int_{\mathcal{B}} p_s dx$: upward force from fluid pressure on the wetted region
- ▶ At equilibrium, $\int_{\mathcal{B}} p_s dx = M/Fr$, i.e., fluid supports body weight

Non-dimensionalisation

Characteristic scales: length L_c , velocity $U = L_c/T_c$.

$$x \rightarrow x/L_c, \quad t \rightarrow t U/L_c, \quad \phi \rightarrow \phi/(UL_c), \quad p \rightarrow p/(\rho U^2)$$

Number	Definition	Physical meaning
Fr	$U^2/(gL_c)$	inertia vs. gravity
We	$\rho U^2 L_c/\sigma$	inertia vs. surface tension
Re	UL_c/ν	inertia vs. viscosity
M	$m/(\rho L_c^2)$	body mass (dimensionless, per unit length)

Default scales in the code: $L_c = 2.5$ cm, $T_c = 0.05$ s $\Rightarrow U = 50$ cm/s.

Default dimensionless parameters:

$$\text{Fr} = 1.0204, \quad \text{We} = 35.7347, \quad \text{Re} = 250\,000$$

Spatial discretisation

Uniform rectangular grid: n_x nodes in x , n_z nodes in z .

$$x_j = j \Delta x, \quad j = 0, \dots, n_x - 1; \quad z_i = -D + i \Delta z, \quad i = 0, \dots, n_z - 1$$

($i = 0$ bottom, $i = n_z - 1$ surface, periodic in x .)

Second-order finite-difference operators (all periodic in x):

- ▶ Horizontal Laplacian: $(\Delta_H u)_j = (u_{j+1} - 2u_j + u_{j-1})/\Delta x^2$
- ▶ Centered first derivative: $(D_x u)_j = (u_{j+1} - u_{j-1})/(2\Delta x)$
- ▶ 2D Laplacian: 5-point stencil with periodic lateral wrapping

Boolean body mask $\chi_j \in \{0, 1\}$: 1 if column j lies under the body.

```
body_mask = np.abs(x - xc_obj) < a_obj (from _setup)
```

The augmented state vector

All unknowns at time t^{n+1} stacked into one vector:

$$\mathbf{x} = \begin{pmatrix} \vec{w}^3 \\ \vec{w}^1 \\ \vec{\phi} \\ \vec{\eta} \\ \vec{p}_s \\ z_b \\ v_b \end{pmatrix}, \quad N_{\text{total}} = 3 n_x n_z + 2 n_x + 2$$

Block	Size	Role
$\vec{w}^3$	$n_x n_z$	vertical vortical component (2D field)
$\vec{w}^1$	$n_x n_z$	horizontal vortical component (2D field)
$\vec{\phi}$	$n_x n_z$	velocity potential (top row = ϕ_s)
$\vec{\eta}$	n_x	surface elevation
$\vec{p}_s$	n_x	surface pressure (zero off body)
z_b, v_b	2	body scalar states

For $n_x = 400, n_z = 80$: $N_{\text{total}} \approx 96,800$ — manageable with sparse LU.

Monolithic matrix: block structure

$$A \mathbf{x}^{n+1} = \mathbf{b}(\mathbf{x}^n)$$

	w^3	w^1	ϕ	η	p_s	z_b	v_b
w^3	✓		★				
w^1	★	✓	★				
ϕ	★		✓		★		
η	★		★	✓		★	
p_s	★		★		✓		★
z_b						✓	★
v_b					★		✓

✓ diagonal block (implicit in that field). ★ coupling block (cross-field dependency).

Key property

All coefficients come from grid, Δt , (Fr, We, Re, M) , and the body mask. **A does NOT depend on the solution** $\Rightarrow$ factor once, reuse forever.

w^3 block: three row types

Interior ($1 \leq i \leq n_z - 2$): heat equation

$$\left(I - \frac{\Delta t}{\text{Re}} \Delta_{2D} \right) w^{3,n+1} = w^{3,n}$$

diagonal $1 + 2\Delta t/\text{Re} (\frac{1}{\Delta x^2} + \frac{1}{\Delta z^2})$, $-\Delta t/\text{Re} \Delta x^{-2}$ on x -neighbours, etc.

Bottom ($i = 0$): implicit non-slip, couples to ϕ

$$w_{0,j}^{3,n+1} + \frac{\phi_{1,j}^{n+1} - \phi_{0,j}^{n+1}}{\Delta z} = 0$$

Surface ($i = n_z - 1$): heat eq inclusively **[v3]**

$$\left(I - \frac{\Delta t}{\text{Re}} \Delta_{2D} \right) w_s^{3,n+1} = w_s^{3,n}$$

with one-sided ∂_{zz} stencil $(w_{-1}^3 - 2w_{-2}^3 + w_{-3}^3)/\Delta z^2$. **v3 update**: the auxiliary surface vorticity equation (2.15e) is dropped. Instead, the bulk heat equation is applied "inclusively" at the surface row,

w^1 block: piecewise surface BC

Interior and bottom identical in form to w^3 (heat eq. + implicit non-slip, coupling to ϕ).

Surface: the equation *switches* based on χ_j .

Free node ($\chi_j = 0$): stress-free (2.11a) [v3 fix]

The full tangential-stress balance $\sigma_{xz} = \mu(\partial_x u_z + \partial_z u_x) = 0$ gives

$$2 \partial_{xz} \phi + \partial_x w^3 + \partial_z w^1 = 0,$$

discretised as

$$w_{-1,j}^{1,n+1} - w_{-2,j}^{1,n+1} + \Delta z (D_x w_{-1}^{3,n+1})_j = -2 \Delta z (\partial_{xz} \phi^n)_j.$$

Factor of 2 on ϕ_{xz} (both $\partial_x u_z$ and $\partial_z u_x$ contribute via the symmetric mixed derivative). v2 had only a single ϕ_{xz} , which was an algebra error.

Body node ($\chi_j = 1$): horizontal no-slip

$$w_{-1,j}^{1,n+1} + (D_x \phi_{-1}^{n+1})_j = 0$$

couples w^1 surface to ϕ surface (implicit).

ϕ block: Laplace + dynamic BC, coupled to w^3, p_s, η

Bottom ($i = 0$): Laplace with ghost-point non-slip

$$(-2\alpha - 2)\phi_{0,j} + \alpha(\phi_{0,j\pm 1}) + 2\phi_{1,j} - 2\Delta z w_{0,j}^3 = 0, \quad \alpha = (\Delta z / \Delta x)^2$$

Interior ($1 \leq i \leq n_z - 2$): 5-point Laplacian

$$(-2\alpha - 2)\phi_{i,j} + \alpha\phi_{i,j\pm 1} + \phi_{i\pm 1,j} = 0$$

Surface ($i = n_z - 1$): dynamic BC (2.15d), uniform form

$$\left(1 - \frac{2\Delta t}{\text{Re}} \Delta_H\right) \phi_{-1,j}^{n+1} + \frac{2\Delta t}{\text{Re}} \frac{w_{-1,j}^3 - w_{-2,j}^3}{\Delta z} + \Delta t p_{s,j}^{n+1} = \phi_{-1,j}^n - \frac{\Delta t}{\text{Fr}} \eta_j^n + \frac{\Delta t}{\text{We}} \kappa[\eta^n]_j$$

Same row at every j — the p_s term is always present. Whether p_s is free or unknown is decided by a *different* row (the p_s block).

η and p_s blocks: the constraint switch

η rows

Free node: eq. (2.15c)

$$\eta_j^{n+1} - \frac{\Delta t}{\Delta z} (\phi_{-1,j}^{n+1} - \phi_{-2,j}^{n+1}) - \Delta t w_{-1,j}^{3,n+1} = \eta_j^n$$

Body node: geometric constraint

$$\eta_j^{n+1} - z_b^{n+1} = 0$$

p_s rows

Free node: pressure pinned

$$p_{s,j}^{n+1} = 0$$

Body node: kinematic match

$$\frac{\phi_{-1,j}^{n+1} - \phi_{-2,j}^{n+1}}{\Delta z} + w_{-1,j}^{3,n+1} - v_b^{n+1} = 0$$

p_s on body = Lagrange multiplier: whatever pressure is needed to make the fluid surface velocity equal v_b .

Body rows: discretised Newton's 2nd law

z_b equation (implicit Euler of $\dot{z}_b = v_b$)

$$z_b^{n+1} - \Delta t v_b^{n+1} = z_b^n$$

v_b equation — couples to integrated p_s

$$M v_b^{n+1} - \Delta t \Delta x \sum_{j \in \mathcal{B}} p_{s,j}^{n+1} = M v_b^n - \frac{\Delta t M}{\text{Fr}}$$

Derivation:

- ▶ Start from $M \dot{v}_b = -M/\text{Fr} + \int_{\mathcal{B}} p_s \, dx$
- ▶ Implicit Euler: $M(v_b^{n+1} - v_b^n)/\Delta t = -M/\text{Fr} + \Delta x \sum_{j \in \mathcal{B}} p_{s,j}^{n+1}$
- ▶ Rearrange, move knowns to RHS.

Midpoint sum is the discrete approximation to $\int_{\mathcal{B}} p_s \, dx$.

Why it all works: constant coefficient matrix

The decisive observation

Every matrix entry depends only on:

- ▶ Grid parameters ($n_x, n_z, \Delta x, \Delta z$)
- ▶ Time step Δt
- ▶ Dimensionless numbers ($\text{Fr}, \text{We}, \text{Re}, M$)
- ▶ Body mask χ (fixed once body geometry is set)

No coefficient depends on the solution.

Consequence: use sparse LU decomposition from SciPy `spilu`:

$$A = LU \quad (\text{computed } \textit{once} \text{ at setup})$$

At each time step:

$$\mathbf{x}^{n+1} = U^{-1}(L^{-1}\mathbf{b}(\mathbf{x}^n)) \quad \text{cost: } \mathcal{O}(N) \text{ back-substitution}$$

Same trick as v1, just a slightly bigger system. The monolithic prefactorisation is the key to making this practical.

Algorithm: one full time step

Setup (once, offline)

```
A, offsets = build_monolithic_matrix_v2(nx, nz, dx, dz, dt,  
                                         Fr, We, Re, M, body_mask)  
LU = splu(A)      # LU factorisation:  $O(N^{\{3/2\}})$  ish, done ONCE
```

Per time step (inline back-substitution)

```
def _step(w3, w1, phi, eta, ps, zb, vb):  
    # 1. Assemble explicit RHS from state at time n  
    b = build_monolithic_rhs_v2(w3, w1, phi, eta, zb, vb,  
                                offsets, dx, dz, dt,  
                                Fr, We, Re, M,  
                                body_mask, Delta_H, Dx)  
  
    # 2. Solve  $A x = b$  by back-substitution (cheap)  
    x = LU.solve(b)      #  $O(N)$   
  
    # 3. Unpack into new state at time n+1  
    return unpack_state(x, offsets)
```

Configuration dataclasses

```
@dataclass
```

```
class PhysicalParams:
```

```
    L_cm: float = 50.0      # domain length (cm)
    D_cm: float = 20.0      # depth (cm)
    g: float = 980.0        # gravity (cm/s^2)
    sigma: float = 174.9    # surface tension (dyne/cm)
    rho: float = 1.0        # density (g/cm^3)
    nu: float = 0.0005      # kinematic viscosity (cm^2/s)
```

```
@dataclass
```

```
class BodyParams:
```

```
    half_width_cm: float = 5.0
    center_x_cm: float = 25.0
    mass_per_length_g_per_cm: float = 0.5 # -> dimensionless M
    initial_z_cm: float = 0.0 # where the body starts
    initial_vz_cm_s: float = 0.0 # initial velocity (negative = drop)
```

```
@dataclass
```

```
class NumericalParams:
```

```
    nx: int = 400; nz: int = 80
    nt: int = 4000; total_time_s: float = 2.0
    unit_length: float = 2.5; unit_time: float = 0.05
```

Matrix builder: body dynamics rows

```
# Row block 6:  $z_b \dot{z}_b = v_b$ 
A[z_b, z_b] = 1.0
A[z_b, v_b] = -dt

# Row block 7:  $v_b$  (Newton's 2nd law)
A[v_b, v_b] = M # inertia
for j in range(nx):
    if body_mask[j]:
        A[v_b, ps(j)] = -dt * dx # integrated pressure coupling

# Row block 5:  $p_s$  (the Lagrange-multiplier rows)
for j in range(nx):
    row = ps(j)
    if body_mask[j]:
        # Kinematic match:  $\phi_{nz-1} + w_3 - v_b = 0$ 
        A[row, phi(nz-1, j)] = 1.0 / dz
        A[row, phi(nz-2, j)] = -1.0 / dz
        A[row, w3(nz-1, j)] = 1.0
        A[row, v_b] = -1.0
    else:
        A[row, ps(j)] = 1.0 # pin to zero
```


Solver class: FloatingObjectSolverV2

```
class FloatingObjectSolverV2:
    def __init__(self, config):
        self.cfg = config
        self._setup()                # grid, masks, operators

        # Build A once, factor once
        A, self.off = build_monolithic_matrix_v2(...)
        self.LU = splu(A)

    def _step(self, w3, w1, phi, eta, ps, zb, vb):
        b = build_monolithic_rhs_v2(
            w3, w1, phi, eta, zb, vb,
            self.off, self.dx, self.dz, self.dt,
            self.Fr, self.We, self.Re, self.M,
            self.body_mask, self.Delta_H, self.Dx,
        )
        return unpack_state(self.LU.solve(b), self.off)

    def run(self):
        # initialise, loop over nt steps, return history dict
        ...
```

Theoretical expectation: hydrostatic equilibrium

At rest, Newton's 2nd law collapses to a force balance:

$$0 = -\frac{M}{\text{Fr}} + \int_{\mathcal{B}} p_s \, dx \quad \Longrightarrow \quad \int_{\mathcal{B}} p_s \, dx = \frac{M}{\text{Fr}}$$

If p_s is (quasi-)uniform under the body:

$$p_s \approx -\frac{\eta}{\text{Fr}} = -\frac{z_b}{\text{Fr}} \quad \Longrightarrow \quad \boxed{|z_b^{\text{eq}}| = \frac{M}{2a} = \frac{m}{\rho \cdot 2a}}$$

Archimedes rediscovered! Draft = mass per unit length $\div$ (fluid density $\times$ body width).

Example: $m = 0.5 \text{ g/cm}$, $2a = 10 \text{ cm}$, water ($\rho = 1 \text{ g/cm}^3$):

$$|z_b^{\text{eq}}| = 0.5 / 10 = 0.05 \text{ cm} = 0.5 \text{ mm}$$

Smoke-test: release from rest

Parameters (small grid for quick iteration):

- ▶ Grid $n_x \times n_z = 80 \times 20$, $\Delta t = 5$ ms, 30 steps ($t_{\text{end}} = 0.15$ s)
- ▶ $m = 0.5$ g/cm, $2a = 10$ cm; $\text{Fr} = 1.02$, $\text{We} = 35.7$, $\text{Re} = 2.5 \times 10^5$
- ▶ Initial condition: $z_b(0) = 0$, $v_b(0) = 0$ (released from rest at undisturbed level)

Solver output:

```
FloatingObjectSolverV2 initialised
```

```
Grid: 80x20, Body: 2a=10.0 cm, M_dim=0.0800
```

```
Body nodes: 15/80
```

```
Monolithic size N = 4962, nnz = 23938
```

```
Matrix build: 0.02s, LU factor: 0.01s
```

```
...
```

```
Step      30 | t=0.150s | z_b=-0.593mm | v_b=-0.317cm/s | F=+7.88e-02
```

Smoke test: sanity checks

Quantity	Expected	Observed (at $t = 0.15$ s)
p_s off body	$= 0$ (machine precision)	$\max p_s^{\text{free}} = 0$
p_s on body	edge-concentrated, positive	$[0.033, 0.021, \dots, 0.018, \dots, 0.033]$
$\int_{\mathcal{B}} p_s \, dx$	$M/\text{Fr} = 0.0784$	0.0788 (+0.5%)
Body motion	sink under gravity, decelerate	$z_b : 0 \rightarrow -0.59$ mm; v_b reversing
Performance	factor once, solve fast	LU 0.01 s; step 0.37 ms

Verdict

All qualitative signatures present: constraint exactly enforced, force converges to Archimedes prediction, body dynamics physical, cost negligible.

The +0.5% in $\int p_s \, dx$ is the acceleration term $M\dot{v}_b$ — body is still decelerating at $t = 0.15$ s.

Another scenario: drop impact

Parameters: $m = 5 \text{ g/cm}$, $z_b(0) = 0$, $v_b(0) = -20 \text{ cm/s}$ (dropping at 20 cm/s).

Expected dynamics:

1. Body plunges below surface (accelerated by gravity, decelerated by fluid reaction)
2. Plunge depth $\sim 7 \text{ mm}$ (rough estimate: kinetic energy $\rightarrow$ potential energy)
3. Rebound: body rises, overshoots equilibrium, oscillates
4. Oscillation decays by wave radiation + viscous dissipation
5. Equilibrium draft: $m/(\rho \cdot 2a) = 5/10 = 5 \text{ mm}$

Oscillation period (theoretical, added-mass estimate):

$$T \approx 2\pi \sqrt{\frac{M_{\text{eff}}}{2a \cdot (\rho g / L_c)}} \quad \text{where } M_{\text{eff}} = m + m_{\text{added}}$$

Added mass for a flat 2D strip on a fluid surface is $\mathcal{O}(\rho a^2)$ — so heavy bodies oscillate slowly, light bodies respond quickly.

Caveat: periodic boundary $\Rightarrow$ wave build-up

Radiated waves wrap around in the periodic domain and **return to the body**, causing spurious interactions.

Symptoms:

- ▶ Wave amplitudes at $x = 0/L$ grow with time
- ▶ Body fails to reach full hydrostatic equilibrium
- ▶ Apparent “energy growth” because confinement prevents radiation

Diagnosis: inspect $\eta(x, t)$ space-time diagram — if outward-propagating wavefronts meet the boundary and re-enter, you’re in the wrap-around regime.

This is a numerical artefact, not a bug in the solver.

Mitigations:

- ▶ **Shorter runs:** $t_{\text{end}} < L/(2c_{\text{wave}})$
- ▶ **Larger domain:** $L \gg c_{\text{wave}} t_{\text{end}}$
- ▶ **Absorbing/sponge layer:** add damping in a strip near boundaries (not yet implemented)

For quantitative single-oscillation physics, use shorter t_{end} or scale up L .

Limitations of v2

- ▶ **Linearised free surface:** $|\eta| \ll D$ assumption. Breaks down for violent impacts or very heavy bodies.
- ▶ **Implicit Euler time stepping:** first-order accurate. Sufficient for stable dynamics, but not for high-accuracy long-time integration.
- ▶ **Flat-bottomed rigid body only:** curved bodies, deformable bodies, and non-rigid contact not supported.
- ▶ **2D Cartesian only:** no axisymmetric, no 3D.
- ▶ **Periodic horizontal BC:** wave wrap-around, no open-boundary radiation.
- ▶ **No contact-line physics:** body edge is a sharp discontinuity; real physics there is more complex.
- ▶ **No separation / reattachment:** once the body contacts the surface, it stays in contact for the whole simulation.

Natural extensions

Time integration

VS-BDF2 for second-order accuracy — the matrix stays constant, only the Euler weights change. [Hairer & Wanner 1996]

Domain and geometry

- ▶ **Absorbing/sponge layer** near periodic boundaries to suppress reflection.
- ▶ **Axisymmetric coordinates** for sphere impact (Galeano-Rios et al. 2017; Agüero et al. 2022).
- ▶ **Multiple bodies** — add more Lagrange-multiplier blocks; matrix remains sparse and constant.

Physics

- ▶ **Curved bodies**: generalise the constraint to $\eta = z_b + s(x - x_c)$.
- ▶ **Faraday forcing**: add vertical-vibration term — pilot-wave analogues (Milewski et al. 2015).
- ▶ **Elastic membrane**: couples to a different BVP, same constraint-multiplier structure.

References

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- ▶ Galeano-Rios (2016). Hydrodynamic Pilot-waves. PhD thesis, IMPA, Rio de Janeiro. *[Additional derivation detail.]*
- ▶ Agüero, Alventosa, Harris, Galeano-Rios (2022). Impact of a rigid sphere onto an elastic membrane. *Proc. R. Soc. A* **478**, 20220340.
- ▶ Lamb, H. (1932). *Hydrodynamics*, 6th ed., Cambridge University Press.
- ▶ Batchelor, G.K. (1967). *An Introduction to Fluid Dynamics*, Cambridge University Press.
- ▶ Hairer & Wanner (1996). *Solving ODEs II: Stiff and DAE Problems*, Springer.

Code: 03_Floating_Object/floating_object_v2.py

Manuscript: 99_overleaf_Floating_LNS_Wongyung/main.tex (§2.5, §3.4)

Summary

What v2 does

A flat-bottomed rigid body floats on a viscous fluid. Its motion is determined by gravity and the self-consistent fluid pressure — no prescribed kinematics.

How it works

Augment the unknowns by (p_s, z_b, v_b) . Replace free-surface BCs with three constraints under the body. Use the surface pressure as a Lagrange multiplier. Assemble one monolithic matrix, factor once, back-solve each step.

Why it works

All equations are linear with coefficients that do not depend on the solution. The $\mathcal{O}(N)$ back-substitution cost per step makes the fully-implicit coupling practical.

Questions?